

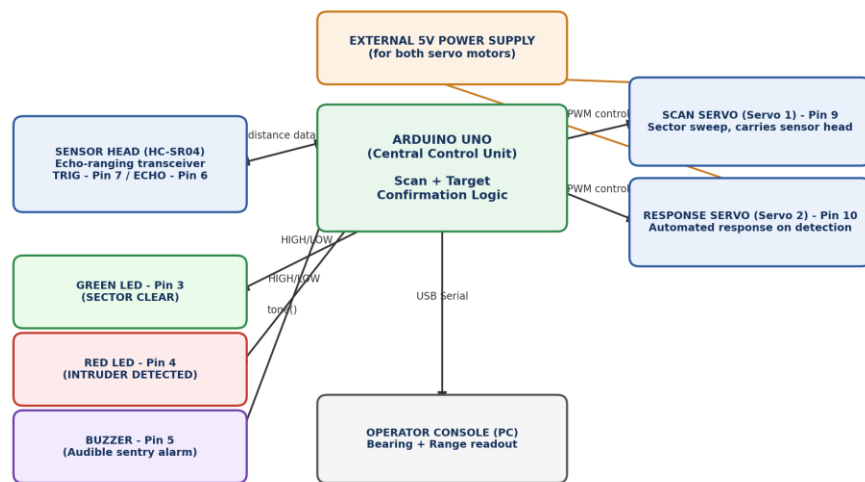
PROJECT REPORT

Automated Border / Perimeter Sentry Alert System

A Radar-Style Intrusion Detection Model for Defence & Border Security Awareness

(Servo-Driven Scanning Detector using Arduino Uno - Educational Prototype)

Fig 1: Block Diagram - Perimeter Sentry Radar Model



Platform: Arduino Uno (ATmega328P)
Language: Arduino C/C++ (Servo Library)
Date: 15 September 2026

1. Introduction

Border outposts and defence perimeters need continuous, reliable watch over the ground in front of them, especially at night or in low-visibility conditions where human sentries alone may miss a slow, quiet approach. This project presents a small-scale educational model of an automated sentry / perimeter-watch system, built around an Arduino Uno microcontroller, that illustrates how a low-cost sensor-and-alarm setup could support a human guard by continuously scanning an area and raising an immediate alert the moment something enters the watched zone.

An ultrasonic distance sensor is mounted on a rotating servo (the “scan servo”) that sweeps back and forth across a 150-degree arc, continuously measuring the distance to whatever lies ahead - similar in principle to how a radar antenna sweeps its beam across a sector. When an object is found consistently within a preset detection range, the system confirms the detection using multiple consecutive readings to reject sensor noise (so a stray echo or leaf blowing past does not cause a false alarm), then raises a visual and audible alarm (red LED and buzzer) and actuates a second “trigger” servo to perform a short automated response action, such as swinging a flag/indicator or triggering a connected alarm relay. When the watched area is clear, a green LED remains lit and the system continues its scan.

This is an educational prototype, not a deployable defence system: it uses a short-range ultrasonic sensor (about 1 metre) and hobby-grade servos purely to demonstrate the underlying concepts - continuous sector scanning, noise-filtered detection, and automated alerting - that are common to real perimeter-surveillance and radar-sentry systems used by armed forces and border security forces. It is intended to build understanding of embedded sensing and automation, and as a scaled-down concept model for how such alert systems work.

2. Objective

The primary objective of this project is to design and build a working table-top demonstration model of a radar-style perimeter surveillance system of the kind conceptually used for border and cantonment security by the Indian Army and other security forces, using low-cost, easily available components. Specific objectives include:

1. To demonstrate the basic working principle of radar - sweeping a beam across a sector and measuring the distance to objects in its path - using an ultrasonic sensor as a scaled-down stand-in for a radar transceiver.
2. To interface an HC-SR04 ultrasonic sensor with an Arduino Uno for accurate short-range distance measurement (up to 1 metre in this model).
3. To use a servo motor to continuously sweep the sensor across a 150-degree field of view, replicating the sector-scan action of a surveillance radar antenna.
4. To implement a noise-filtering algorithm that requires several consistent readings before confirming a genuine detection - illustrating how real systems avoid false alarms from birds, foliage, or stray echoes.
5. To provide immediate sentry-style status indication using LEDs (green = sector clear, red = intruder detected) and an audible buzzer alarm.
6. To automatically actuate a secondary servo on confirmed detection, representing an automated response action (alarm relay, indicator flag, or camera/searchlight slewing toward the target).
7. To log scan angle and target distance to a PC over the serial port, representing the operator's radar console display.

8. To serve as an educational aid for understanding defence surveillance technology, embedded sensing, and automation concepts.

3. Block Diagram

The figure below shows the overall system architecture, laid out in the same functional blocks found in a real surveillance radar installation: a scanning sensor head, a central processing/control unit, an operator display console, and an alarm & response sub-system. The Arduino Uno acts as the central controller: it reads target distance data from the ultrasonic sensor head, positions the scan servo, drives the sentry status indicators (LEDs and buzzer), actuates the response servo, and reports scan data over USB serial to the operator console. The two servos are powered from an independent external 5V supply (with a common ground to the Arduino) since the Uno's onboard 5V rail cannot safely supply two servo motors.

Fig 1: Block Diagram - Perimeter Sentry Radar Model

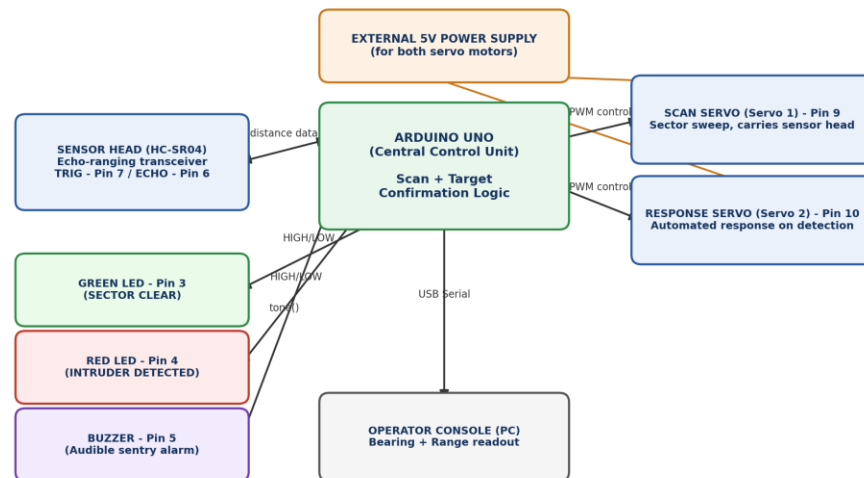


Figure 1: Block diagram of the ultrasonic sentry / radar detection system

4. Components Used

All components are low-cost, readily available modules chosen to keep the demonstration model affordable and easy to reproduce in a college laboratory.

Component	Quantity	Function in Project
Arduino Uno	1	Main controller; runs the detection & scanning logic
HC-SR04 Ultrasonic Sensor	1	Measures distance to nearby objects (range up to 100 cm)
SG90 / MG90 Servo Motor	2	Servo 1 sweeps the sensor 0-180 deg; Servo 2 performs the trigger action
Green LED	1	Indicates area is clear

Component	Quantity	Function in Project
Red LED	1	Indicates an object/target has been detected
Buzzer	1	Sounds an audible alert on detection
220 ohm Resistors	2	Current-limiting resistors for the LEDs
External 5V Power Supply	1	Powers both servos independently of the Arduino's onboard 5V
Jumper Wires & Breadboard	-	Circuit connections

5. Pin Diagram / Circuit Connections

The connection diagram below shows how each component wires to the Arduino Uno's digital pins, and how power/ground is distributed. All grounds - Arduino, ultrasonic sensor, LEDs, buzzer, and the external servo power supply - must be tied together (common ground) for the circuit to function correctly.

Fig 2: Pin Connection Diagram - Perimeter Sentry Radar Model

* External 5V supply GND must be common with Arduino GND

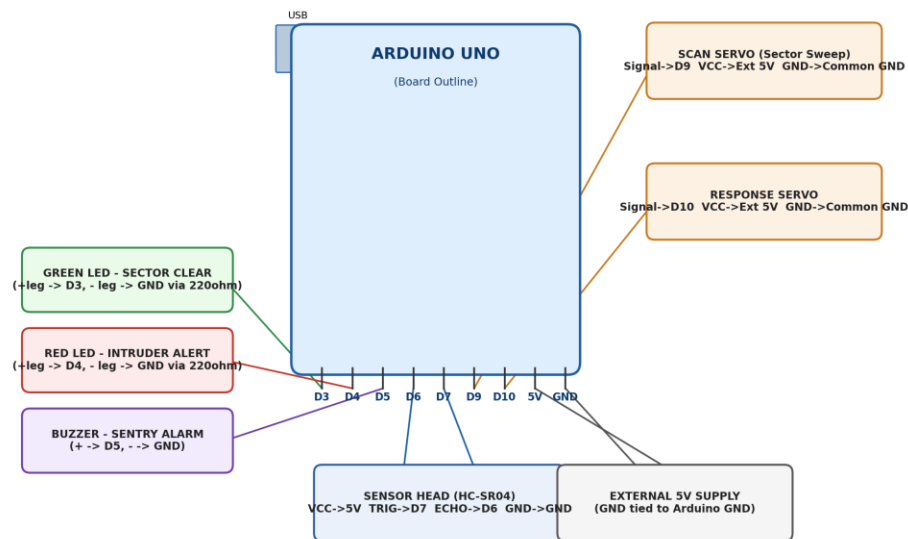


Figure 2: Pin connection diagram

5.1 Pin Connection Table

Component	Component Pin	Arduino Uno Pin
Scan Servo (Servo 1)	Signal	Digital Pin 9 (PWM)
Trigger Servo (Servo 2)	Signal	Digital Pin 10 (PWM)
Ultrasonic Sensor HC-SR04	TRIG	Digital Pin 7 (Output)
Ultrasonic Sensor HC-SR04	ECHO	Digital Pin 6 (Input)

Component	Component Pin	Arduino Uno Pin
Green LED	Anode (+)	Digital Pin 3 (Output)
Red LED	Anode (+)	Digital Pin 4 (Output)
Buzzer	Positive (+)	Digital Pin 5 (Output)
Ultrasonic Sensor VCC / Servos VCC	VCC	5V (sensor: Arduino 5V; servos: external 5V supply)
All modules	GND	Common GND (Arduino + external supply)

Note: Both servo motors should be powered from the external 5V supply and NOT from the Arduino's onboard 5V pin, as two servos can draw more current than the Uno's regulator can safely provide.

6. Working Principle / Methodology

1. **SYSTEM INITIALISATION:** On power-up, the Arduino initialises the serial console link, sets all pin modes, attaches both servo objects, returns the response servo to its stowed position, and lights the green LED - indicating the sector is clear and the sentry is on watch.
2. **SECTOR SCANNING:** In the main loop, the scan servo is stepped one degree at a time between 15 deg and 165 deg (avoiding the mechanical end-stops), reversing direction at each limit. This produces the continuous back-and-forth sector sweep characteristic of a surveillance radar antenna.
3. **RANGING (ECHO PRINCIPLE):** At every scan bearing, the ultrasonic sensor transmits a 10-microsecond pulse and measures the echo return time using `pulseIn()`. Since the pulse travels out and back, the range is calculated as $(\text{time} \times \text{speed of sound}) / 2$. This is the same time-of-flight principle real radar uses with radio waves instead of sound. Returns beyond 100 cm or closer than 8 cm are gated out as invalid.
4. **TARGET CONFIRMATION:** If a single return falls within the 20 cm detection threshold, the system does NOT raise an alarm immediately. It takes 6 consecutive samples, and only if ALL 6 confirm a target in range is the detection declared valid. This mirrors the multi-hit confirmation logic used in real systems to reject clutter, birds, and stray echoes before alerting an operator.
5. **ALARM & AUTOMATED RESPONSE:** On a confirmed intrusion, the red LED and buzzer activate (sentry alarm), the confirmed target range is transmitted to the operator console, and the response servo swings to 90 degrees and back - representing an automated response action such as slewing a camera or searchlight onto the target, or tripping an alarm relay. The system then reverts to the clear state and resumes its scan.
6. **OPERATOR CONSOLE DISPLAY:** Scan bearing (angle) and target range are continuously streamed over serial to the PC, giving the operator a live readout of the sector - the model's equivalent of a radar console.

7. Key Firmware Features

- **Noise rejection:** an object is only confirmed after 6 consecutive matching readings (`confirmSamples`), avoiding false triggers from sensor jitter.
- **Bounded sweep:** the scan angle is limited to 15 deg-165 deg to protect the servo from stalling at its mechanical limits.

- Range gating: readings below 8 cm or above 100 cm are discarded as invalid, keeping the sensing window realistic and reliable.
- Non-blocking status indication: LEDs and buzzer are updated every loop iteration, giving immediate visual feedback of system state.
- Serial diagnostics: live angle and distance values are streamed over USB for testing and calibration.

8. Applications

8.1 Defence & Security Concepts Demonstrated

The following are the real-world defence and border-security applications whose operating principles this model demonstrates. The prototype itself is a short-range educational model; actual deployed systems use radar, thermal imaging, and seismic sensors over far greater ranges.

- Border and Line of Control (LoC) perimeter surveillance - automated watch over a sector to detect infiltration attempts, supporting rather than replacing human sentries.
- Cantonment, ammunition depot, and airbase perimeter intrusion detection.
- Unattended ground sensor posts for areas that are difficult or unsafe to man continuously, particularly at night or in fog and low visibility.
- Automated slewing of surveillance cameras, searchlights, or thermal imagers onto a detected contact.
- Forward post early-warning alarms that alert a guard room the moment a sector is breached.
- Battlefield obstacle detection and navigation assistance for unmanned ground vehicles and robotic platforms.

8.2 Civil and Educational Applications

- Intrusion monitoring for homes, offices, warehouses, and restricted rooms.
- Railway crossing and tunnel obstacle-detection alerts.
- Parking assistance and blind-spot proximity warning in vehicles.
- Industrial safety zones - alerting when personnel enter a hazardous machine area.
- Educational demonstration of radar/echo-ranging principles, PWM servo control, and sensor noise-filtering techniques in engineering curricula.

9. Advantages and Limitations

9.1 Advantages

- Low-cost components and simple, widely available Arduino hardware.
- Wide 150-degree field of coverage from a single sensor via mechanical scanning.
- Built-in software filtering reduces false alarms caused by sensor noise.
- Clear, simple status feedback (LED + buzzer) that is easy to observe.

9.2 Limitations

It is important to state clearly what separates this demonstration model from an operational surveillance system:

- Range: ultrasonic sensing is limited to about 1 metre here, whereas actual border surveillance radars operate over hundreds of metres to several kilometres.
- Medium: sound waves are heavily affected by wind, temperature, humidity, and background noise, and are absorbed by soft or angled surfaces. Real systems use radio waves (radar) or infrared/thermal imaging, which are far more robust outdoors.
- Scan rate: mechanical sweeping is slow compared to electronically-steered (phased array) radar, so a fast-moving target could pass between sweeps.
- Field of view: a single sensor looks in only one direction at a time; operational installations use multiple overlapping sensors for full, gap-free coverage.
- No classification: the model only detects presence, it cannot distinguish a person from an animal or vehicle. Real systems use imaging and signature analysis for target classification.
- Power and ruggedisation: hobby components are not weatherproof, and would require environmental sealing, surge protection, and reliable field power for any outdoor use.

10. Conclusion

This project successfully demonstrates a working table-top model of a radar-style perimeter surveillance system, of the type conceptually employed for border and installation security. By combining ultrasonic echo-ranging, servo-driven sector scanning, and noise-filtered detection logic on an Arduino Uno, the model reproduces the essential functional chain of a real surveillance installation: scan the sector, range the contact, confirm the target, alert the operator, and initiate an automated response.

All stated objectives were met. The system sweeps a 150-degree sector continuously, measures range accurately within its rated distance, and - importantly - uses six-sample confirmation logic to reject spurious returns, making it markedly more dependable than a naive single-reading threshold design. Green and red indicators plus an audible buzzer give instant sentry-style status, while the serial console provides a live bearing-and-range readout analogous to an operator display.

The project also makes clear the gap between a demonstration model and a deployable system. Real border surveillance relies on radar, thermal imaging, and networked unattended ground sensors operating over far greater ranges and in harsh weather, with target classification and command-post integration. What this model conveys is the underlying engineering logic common to all of them. As an educational exercise it builds practical understanding of sensor interfacing, PWM actuator control, real-time decision-making, and false-alarm mitigation - the foundational skills behind such defence and security technologies - and provides a platform that can be extended toward wireless alerting, data logging, and multi-sensor coverage.

11. Future Scope

- Add a Processing/Python PPI radar console on the PC to display the live sweep and plotted contacts, closely resembling an actual radar operator screen.
- Replace the ultrasonic sensor with an IR, LiDAR, or mmWave radar module to extend detection range substantially and improve outdoor reliability.

- Integrate a PIR or thermal sensor alongside the ranging sensor so the system can distinguish a human/warm body from an inanimate object.
- Add Wi-Fi/GSM connectivity (ESP32 or SIM module) to relay intrusion alerts instantly to a remote guard room or a sentry's mobile phone.
- Mount a camera module that automatically slews to the detected bearing and captures an image for visual confirmation.
- Deploy multiple scanning nodes with overlapping sectors, networked to a single command console for continuous gap-free perimeter coverage.
- Log every detection event with bearing, range, and timestamp to an SD card or server for post-incident analysis.
- Add solar power and weatherproof enclosure to allow unattended field operation.